

EduPulse AI

Machine Learning Training Results

1. Dataset Loaded Successfully

Property	Value
Dataset File Path	data/procrastination_dataset.csv
Dataset Shape	100000 rows, 13 columns

First 5 Records:

study_hours_per_day	app_usage_minutes	idle_time_minutes	lms_logins_per_week	submission_offset_hours	completion_rate_percent	deadline_misses_30d	streak_days	avg_session_length_min	distraction_visits_per_day	sleep_hours	risk_level	is_procrastinator
2.83	115.3	27.6	9	20.7	48.9	9	8	33.5	11	6.11	Moderate	1
3.12	124.9	53.7	9	19.4	72.6	6	3	30.2	9	6.23	Moderate	1
2.86	129.9	38.0	10	-1.5	39.3	7	8	36.0	9	7.18	Moderate	0
5.59	147.9	40.5	10	40.9	73.2	1	8	54.5	5	6.42	Low	0
1.13	269.7	57.5	10	6.4	36.1	12	0	25.3	16	5.19	High	1

2. Preparing Features and Target

Property	Details
Target Column	is_procrastinator
Features Shape	100000 rows, 11 columns
Target Shape	100000 samples

Feature List:

Feature Name	Index
[+] study_hours_per_day	Feature 1
[+] app_usage_minutes	Feature 2
[+] idle_time_minutes	Feature 3
[+] lms_logins_per_week	Feature 4
[+] submission_offset_hours	Feature 5
[+] completion_rate_percent	Feature 6
[+] deadline_misses_30d	Feature 7
[+] streak_days	Feature 8
[+] avg_session_length_min	Feature 9
[+] distraction_visits_per_day	Feature 10
[+] sleep_hours	Feature 11

3. Train/Test Split

Split Property	Value
Split Ratio	80% Train / 20% Test (stratified)
Training Samples	80,000
Testing Samples	20,000
X_train Shape	(80000, 11)
X_test Shape	(20000, 11)
y_train Shape	(80000,)
y_test Shape	(20000,)

4. Feature Scaling

Scaling Attribute	Status / Dimensions
Scaler Algorithm	StandardScaler (Zero mean, Unit variance)
Scaling Status	Scaling Completed Successfully
Scaled Training Shape	(80000, 11)
Scaled Testing Shape	(20000, 11)

5. Logistic Regression Results

Metric Name	Score / Count
Accuracy	0.8188
Precision	0.8212
Recall	0.8168
Specificity	0.8208
F1 Score	0.8190
ROC AUC	0.9034
Average Precision	0.9057
MCC	0.6376
True Positives (TP)	8,200
True Negatives (TN)	8,176
False Positives (FP)	1,785
False Negatives (FN)	1,839

6. Decision Tree Results

Metric Name	Score / Count
Accuracy	0.7334
Precision	0.7330
Recall	0.7376

Specificity	0.7291
F1 Score	0.7353
ROC AUC	0.7334
Average Precision	0.6723
MCC	0.4668
True Positives (TP)	7,405
True Negatives (TN)	7,263
False Positives (FP)	2,698
False Negatives (FN)	2,634

7. Random Forest Results

Metric Name	Score / Count
Accuracy	0.8148
Precision	0.8214
Recall	0.8064
Specificity	0.8233
F1 Score	0.8138
ROC AUC	0.8978
Average Precision	0.8977
MCC	0.6297
True Positives (TP)	8,095
True Negatives (TN)	8,201
False Positives (FP)	1,760
False Negatives (FN)	1,944

8. XGBoost / Gradient Boosting Results

Metric Name	Score / Count
Accuracy	0.8175
Precision	0.8210
Recall	0.8137
Specificity	0.8212
F1 Score	0.8173
ROC AUC	0.9024
Average Precision	0.9048
MCC	0.6349
True Positives (TP)	8,169
True Negatives (TN)	8,180
False Positives (FP)	1,781
False Negatives (FN)	1,870

9. Model Comparison Table

Model	Accuracy	Precision	Recall	Specificity	F1	ROC AUC	Avg Prec	MCC
Logistic Regression	0.8188	0.8212	0.8168	0.8208	0.8190	0.9034	0.9057	0.6376
Decision Tree	0.7334	0.7330	0.7376	0.7291	0.7353	0.7334	0.6723	0.4668
Random Forest	0.8148	0.8214	0.8064	0.8233	0.8138	0.8978	0.8977	0.6297
Gradient Boosting	0.8175	0.8210	0.8137	0.8212	0.8173	0.9024	0.9048	0.6349

10. Best Model Summary

Best Model Attribute	Value
Best Model Name	Logistic Regression
Best Accuracy	0.8188
Best F1 Score	0.8190
Best MCC	0.6376
Training Date	2026-08-08 01:21:52
Dataset Size	100,000 total samples
Total Execution Time	16.69 seconds

11. Files Generated

Artifact Name	Target File Path	Status
best_model.pkl	models/best_model.pkl	✓ Saved
scaler.pkl	models/scaler.pkl	✓ Saved
model_metadata.json	models/model_metadata.json	✓ Saved
train.csv	data/train.csv	✓ Saved
test.csv	data/test.csv	✓ Saved
procrastination_result.pdf	reports/procrastination_result.pdf	✓ Generated